Higher School of Computer and Digital Sciences and Technologies
file structures and data structures
2 nd year secondary cycle

Tutorial series 3

Exercise 1:

1. Describe how a new example is classified in a trained decision tree.
2. Bilal trained a decision tree and noticed that its performance is better on the training set than on the test set. Should he increase or reduce the depth of his tree?
3. Why is Naive Bayes called "naive" ?

Exercise 2:

We want to learn a model to determine whether a customer is interested in buying a certain product (yes or no) based on their gender (Male or Female), age (<18, 18–35, or >35), marital status (Single or Married), and income (Low, Medium, or High).

Given the following sample of training examples:

ID	Gender	Age	Marital Status	Income	Purchase
1	Male	18–35	Married	Medium	No
2	Male	<18	Single	Low	No
3	Male	>35	Married	High	Yes
4	Female	<18	Single	Medium	No
5	Male	18–35	Single	Medium	No
6	Female	18–35	Single	High	Yes
7	Female	18–35	Married	Low	No
8	Male	18–35	Married	High	Yes
9	Male	>35	Single	Low	Yes
10	Female	<18	Single	Medium	No
11	Female	>35	Single	Medium	Yes
12	Female	>35	Married	High	Yes
13	Male	18–35	Single	Low	No
14	Female	18–35	Married	Medium	Yes

1. Build the decision tree from these examples using information gain. Assume we stop subdividing only when the nodes are pure.
2. Does this method guarantee finding the optimal solution in terms of tree complexity

Exercise 3:

The following table contains data on results obtained by first-year university students ("Common Core" students). Each student is described by 3 attributes: whether they are repeating the year, their high school Baccalaureate stream, and their grade. Students are divided into two classes: Admitted and Not Admitted. We want to build a decision tree from the table to determine the factors influencing student success in the Common Core.

#	Repeating	Stream	Grade	Class
1	No	Maths	ABien	Admitted
2	No	Technical	ABien	Admitted
3	Yes	Science	ABien	NotAdmit
4	Yes	Science	Bien	Admitted
5	No	Maths	Bien	Admitted
6	No	Technical	Bien	Admitted
7	Yes	Science	Passable	NotAdmit
8	Yes	Maths	Passable	NotAdmit
9	Yes	Technical	Passable	NotAdmit
10	Yes	Maths	TBien	Admitted
11	Yes	Technical	TBien	Admitted
12	No	Science	TBien	Admitted

Tasks:

- Use the data in the table to build a decision tree using Gini impurty
- Predict results for the following test data:

#	Repeating	Stream	Grade	Class
1	Yes	Maths	Bien	Admitted
2	No	Science	ABien	NotAdmit
3	No	Maths	TBien	Admitted
4	No	Maths	Passable	NotAdmit

Exercise 3:

Given the following table:

Num	Age	Income	Student	Credit Rating	Buys Computer
1	<=30	High	No	Fair	No
2	<=30	High	No	Excellent	No

Num	Age	Income	Student	Credit Rating	Buys Computer
3	31–40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No
7	31–40	Low	Yes	Excellent	Yes
8	<=30	Medium	No	Fair	No
9	<=30	Low	Yes	Fair	Yes
10	>40	Medium	Yes	Fair	Yes
11	<=30	Medium	Yes	Excellent	Yes
12	31–40	Medium	No	Excellent	Yes
13	31–40	High	Yes	Fair	Yes
14	>40	Medium	No	Excellent	No

We want to **predict whether a user will buy a computer** based on the following information:

- **Age** = <=30
- **Income** = Medium
- **Student** = Yes
- **Credit Rating** = Fair

Use a **Naive Bayes classifier** to classify this user.